CodeGate Security Scan Report

Target: examples\vulnerable_app

Scanned: 2026-10-04T16:06:47.225719+00:00

CodeGate version: 0.2.0

Ranker: R2 | Corpus: real-2040-sym | Verifier: not used

11 finding(s) across 14 scanned function(s), 0 file(s) skipped.

Executive Summary

14 function(s) scanned. 6 auto-approved under the certificate. 8 escalated for review, of which 0 were sent to the LLM verifier this run and 6 were resolved without one (auto-approved or already cached).

Severity	Count
ERROR	2
REVIEW	8
WARNING	1

Detection Attribution

Source	Count	What it means
Semgrep (static rules)	3	Deterministic pattern match, zero LLM cost
ML gate (escalated)	8	Model flagged as not-certifiably-safe
LLM verifier (confirmed)	0	Of the escalated units, confirmed vulnerable
Needs manual review	8	Escalated, not resolved by the verifier
Auto-approved	6	Cleared under the certified miss-rate <= alpha

Cost & Token Accounting

Metric	Value
Tokens saved (est.)	228
Tokens spent (est.)	0
LLM calls avoided	6
LLM calls made	0
Estimated cost	$0.0000
Price used	$0.59 / 1M tokens

All figures on this page are estimates. Token counts use a chars/4 heuristic (no tiktoken dependency), not an exact tokenizer count; "tokens saved" is measured against a baseline of sending every scanned function to the LLM verifier exactly once. See docs/ENGINEERING_LOG.md for the full methodology.

Findings Detail

examples\vulnerable_app\app.py:8 -- handle_login()

UNSPECIFIED (Needs manual review) -- severity REVIEW, source model, confidence 0.32

Not certifiably safe under the current calibration -- needs LLM verification.

Excerpt (from line 5):

```
from system_tasks import ping_host


def handle_login(username):
```

```
    conn = seed_demo_data(sqlite3.connect(":memory:"))
    return get_user_by_name(conn, username)
```

examples\vulnerable_app\app.py:13 -- handle_diagnostics()

UNSPECIFIED (Needs manual review) -- severity REVIEW, source model, confidence 0.26

Not certifiably safe under the current calibration -- needs LLM verification.

Excerpt (from line 10):

```
    return get_user_by_name(conn, username)


def handle_diagnostics(host):
    return ping_host(host)
```

examples\vulnerable_app\config.py:5 -- <module>()

CWE-798 (Hardcoded Credentials) -- severity WARNING, source semgrep, confidence 1.00

String literal assigned to a variable whose name looks like a credential (password/secret/token/api_key/access_key).

Excerpt (from line 2):

```
# Vulnerable: hardcoded credential literal (CWE-798) -- ships with every
# clone of this repo and can't be rotated without a code change.
DATABASE_PASSWORD = "Sup3rSecretPassw0rd!"

APP_NAME = "vulnerable_app_demo"  # Safe: not a credential.
```

Remediation:

A credential literal in source code ships with every clone of the repository and every deployed artifact -- it is readable by anyone with source access, persists in version-control history even after being removed, and cannot be rotated without a code change and redeploy. Loading it from an environment variable or a secret store keeps it out of source entirely and lets it be rotated independently of the code.

Before:

```
password = "Sup3rSecretPassw0rd!"
```

After:

```
password = os.environ["DB_PASSWORD"]
```

examples\vulnerable_app\database.py:4 -- get_user_by_name()

UNSPECIFIED (Needs manual review) -- severity REVIEW, source model, confidence 0.33

Not certifiably safe under the current calibration -- needs LLM verification.

Excerpt (from line 1):

```
import sqlite3


def get_user_by_name(conn, username):
    """Vulnerable: SQL query built by string concatenation (CWE-89)."""
    query = "SELECT * FROM users WHERE username = '" + username + "'"
    return conn.execute(query).fetchone()
```

examples\vulnerable_app\database.py:6 -- get_user_by_name()

CWE-89 (SQL Injection) -- severity ERROR, source semgrep, confidence 1.00

SQL query text built by concatenation/formatting with a variable, then executed. Use a parameterized query instead.

Excerpt (from line 3):

```
def get_user_by_name(conn, username):
    """Vulnerable: SQL query built by string concatenation (CWE-89)."""
    query = "SELECT * FROM users WHERE username = '" + username + "'"
    return conn.execute(query).fetchone()
```

Remediation:

Building a SQL query by concatenating or formatting untrusted input directly into the query text lets an attacker change the query's structure -- for example closing a string literal early to inject their own WHERE clause or UNION SELECT. A parameterized query keeps the query's structure fixed and passes values through a separate channel the database driver escapes correctly, so no input can change what the query does.

Before:

```
query = "SELECT * FROM users WHERE username = '" + username + "'"
cur = conn.execute(query)
```

After:

```
cur = conn.execute(
    "SELECT * FROM users WHERE username = ?", (username,)
)
```

examples\vulnerable_app\database.py:15 -- list_active_users()

UNSPECIFIED (Needs manual review) -- severity REVIEW, source model, confidence 0.26

Not certifiably safe under the current calibration -- needs LLM verification.

Excerpt (from line 12):

```
    return conn.execute("SELECT * FROM users WHERE id = ?", (user_id,)).fetchone()


def list_active_users(conn):
    """Safe: no external input reaches the query at all."""
    return conn.execute("SELECT * FROM users WHERE active = 1").fetchall()
```

examples\vulnerable_app\database.py:20 -- seed_demo_data()

UNSPECIFIED (Needs manual review) -- severity REVIEW, source model, confidence 0.35

Not certifiably safe under the current calibration -- needs LLM verification.

Excerpt (from line 17):

```
    return conn.execute("SELECT * FROM users WHERE active = 1").fetchall()


def seed_demo_data(conn):
    conn.execute("CREATE TABLE users (id INTEGER PRIMARY KEY, username TEXT, active INTEGER)")
    conn.execute("INSERT INTO users VALUES (1, 'alice', 1)")
    conn.commit()
```

examples\vulnerable_app\report_utils.py:6 -- slugify()

UNSPECIFIED (Needs manual review) -- severity REVIEW, source model, confidence 0.24

Not certifiably safe under the current calibration -- needs LLM verification.

Excerpt (from line 3):

```
    return celsius * 9 / 5 + 32


def slugify(text):
    """Safe: string formatting only."""
    return text.lower().strip().replace(" ", "-")
```

examples\vulnerable_app\system_tasks.py:5 -- ping_host()

UNSPECIFIED (Needs manual review) -- severity REVIEW, source model, confidence 0.30

Not certifiably safe under the current calibration -- needs LLM verification.

Excerpt (from line 2):

```
import subprocess


def ping_host(host):
    """Vulnerable: shell=True lets shell metacharacters in `host` run
    arbitrary commands (CWE-78) -- e.g. host="1; rm -rf ~"."""
    return subprocess.run(f"ping -c 1 {host}", shell=True, capture_output=True)
```

examples\vulnerable_app\system_tasks.py:8 -- ping_host()

CWE-78 (OS Command Injection) -- severity ERROR, source semgrep, confidence 1.00

Subprocess invoked with shell=True (or os.system/os.popen). If any argument is influenced by external input this is command injection.

Excerpt (from line 5):

```
def ping_host(host):
    """Vulnerable: shell=True lets shell metacharacters in `host` run
    arbitrary commands (CWE-78) -- e.g. host="1; rm -rf ~"."""
    return subprocess.run(f"ping -c 1 {host}", shell=True, capture_output=True)


def ping_host_safely(host):
```

Remediation:

shell=True (or os.system/os.popen) runs the command through a shell, which interprets metacharacters -- ;, |, &&, $(), backticks -- in any input that reaches the command string. Untrusted input can then chain or substitute arbitrary commands. Passing the command as a list of arguments with shell=False invokes the program directly, with no shell present to interpret metacharacters at all.

Before:

```
subprocess.run(f"ping -c 1 {host}", shell=True)
```

After:

```
subprocess.run(["ping", "-c", "1", host], shell=False)
```

examples\vulnerable_app\system_tasks.py:16 -- list_directory()

UNSPECIFIED (Needs manual review) -- severity REVIEW, source model, confidence 0.30

Not certifiably safe under the current calibration -- needs LLM verification.

Excerpt (from line 13):

```
    return subprocess.run(["ping", "-c", "1", host], shell=False, capture_output=True)


def list_directory(path):
    """Safe: no subprocess or shell at all."""
    return os.listdir(path)
```

Coverage & Certificate

Field	Value
Ranker	R2
Corpus version	real-2040-sym
Provisional	no
t_hat (auto-approve threshold)	0.7763
Certified max miss-rate (alpha)	10.00%
Confidence (1 - delta)	95.00%
Calibration vulnerable samples	206
Created	2026-10-04T15:15:07.290441+00:00

What this guarantees: for any function scored at or above t_hat, the probability it is actually vulnerable is at most 10.00%, with 95.00% confidence -- a distribution-free certificate (Learn-Then-Test conformal risk control), not a heuristic claim. It does not guarantee anything about a function scored below t_hat (escalated) beyond "not certified safe."

Methodology & Limitations

CodeGate is a certified-miss-rate security scanner for Python. It escalates to an LLM only what it cannot statistically prove is safe, using Learn-Then-Test conformal risk control over a ranker trained on labeled generated code.

Scope: Python only. Three CWEs are tracked -- CWE-89 (SQL injection), CWE-78 (OS command injection), CWE-798 (hardcoded credentials). Analysis is intraprocedural (one function at a time); cross-function data flow is not modeled. The threat model is benign-accident, not adversarial -- the scanner never executes the code it scans.

Static findings (semgrep) are deterministic pattern matches against a hand-written local ruleset. Model findings come from a ranker scored against a certified threshold; unresolved model findings needed LLM verification that either wasn't attempted (--no-llm or no API key) or didn't return a usable verdict.